

Automated Glaucoma Report Generation via Dual-Attention Semantic Parallel-LSTM and Multimodal Clinical Data Integration

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Abstract—Generative AI for automated glaucoma diagnostic report generation faces two predominant challenges: content redundancy in narrative outputs and inadequate highlighting of pathologically significant features including optic disc cupping, retinal nerve fiber layer defects, and visual field abnormalities. These limitations primarily stem from current multimodal architectures' insufficient capacity to extract discriminative structural-textural patterns from fundus imaging data while maintaining precise semantic alignment with domain-specific terminology in comprehensive clinical reports. To overcome these constraints, we present the Dual-Attention Semantic Parallel-LSTM Network (DA-SPL), an advanced multimodal generation framework that synergistically processes both fundus imaging and supplementary visual inputs. DA-SPL employs an Encoder-Decoder structure augmented with the novel joint dual-attention mechanism in the encoder for cross-modal feature refinement, the parallelized LSTM decoder architecture for enhanced temporal-semantic consistency, and the specialized label enhancement module for accurate disease-relevant term generation. Rigorous evaluation on standard glaucoma datasets demonstrates DA-SPL's consistent superiority over state-of-the-art models across quantitative metrics. DA-SPL exhibits exceptional capability in extracting subtle pathological indicators from multimodal inputs while generating diagnostically precise reports that exhibit strong concordance with clinical expert annotations. Code is here: <https://github.com/CH-YellowOrange/DA-SPL>.

Index Terms—Fundus Imaging, Image Caption, Medical Report Generation, Attention Mechanism, Multimodal Model, Generative AI

I. INTRODUCTION

Glaucoma, a progressive optic neuropathy, remains one of the leading causes of irreversible blindness worldwide [33], [43], [44]. The disease is characterized by gradual loss of retinal ganglion cells, structural deformation of the optic nerve head, and corresponding visual field defects [3], [26]. Because early glaucomatous damage often develops silently and progresses without noticeable symptoms until late stages, a substantial proportion of patients remain undiagnosed or receive treatment only after irreversible vision loss has occurred [23], [25]. Consequently, early detection, precise staging, and continuous monitoring are critical for preventing visual impairment and guiding clinical management.

Recent advances in artificial intelligence (AI) have greatly facilitated the automated analysis of ocular images, particularly color fundus photographs and optical coherence tomography (OCT) scans, for glaucoma screening [33], segmentation of retinal structures [43], and progression prediction [32]. Despite these achievements, most existing approaches are primarily computer vision oriented, focusing on pattern recognition and pixel-level inference tasks [6], [9], [14], [20], [21], [52]. Such methods often operate as black-box systems that yield quantitative indices or segmentation maps but fail to convey clinically interpretable reasoning or integrate domain-specific ophthalmic knowledge into diagnostic narratives. This lack of semantic alignment with expert reasoning limits clinical adoption, as ophthalmologists depend not only on image features but also on contextual cues such as patient history, anatomical variations, and inter-eye asymmetry.

Therefore, there is an urgent need for AI frameworks that go beyond visual analysis toward generating diagnostically coherent and knowledge-driven reports. Such systems should bridge the gap between imaging biomarkers and textual medical interpretation, providing structured outputs that mirror the logical reasoning of human experts. Developing these multimodal, knowledge-integrated models would represent a crucial step toward trustworthy, explainable, and clinically actionable glaucoma assessment.

To bridge this critical gap, we propose the **Dual-Attention Semantic Parallel-LSTM Network (DA-SPL)**, a novel framework that synergizes visual feature extraction with semantic understanding to transform image-based analysis into clinically interpretable diagnostic reports. By integrating advanced computer vision and natural language processing techniques, DA-SPL generates comprehensive reports that maintain both diagnostic accuracy and clinical relevance.

The principal contributions of this work are three-fold:

- We introduce DA-SPL, an innovative glaucoma report generation framework that effectively combines multimodal visual features with semantic understanding through a dual-attention encoder and parallel LSTM decoder architecture, enhanced by a specialized label refinement module for improved diagnostic precision.

- We propose a novel adaptive weight balancing mechanism within DA-SPL that optimizes feature representation learning by dynamically adjusting to the intrinsic statistical properties of the dataset, resulting in more accurate and clinically relevant report generation.
- Through extensive experimentation on benchmark glaucoma datasets, we demonstrate the superior performance of our proposed architecture and provide comprehensive ablation studies validating the contribution of each component.

II. RELATED WORK

A. AI For Glaucoma

Over the past decade, a variety of publicly available datasets have been introduced to accelerate research at the intersection of artificial intelligence and glaucoma diagnosis. Most existing resources primarily focus on computer vision tasks using retinal fundus or OCT images. For instance, the LAG dataset [25] contains 5,824 fundus images, serving as a benchmark for automated glaucoma detection. Similarly, OCTA-500 [27] provides optical coherence tomography angiography (OCTA) images from 500 subjects, supporting cross-modality vascular analysis. Large-scale resources such as Harvard-GDP [34] and Harvard-GF [35] include 21,059 and 3,300 retinal samples, respectively, enabling population-level investigations into glaucomatous features. Several mid-sized datasets have contributed to optic disc and cup segmentation, including FairSeg [43], FairDomain [44], and classical collections such as Drishti-GS [41], ORIGA [52], and RIM-ONE [9]. Foundational retinal datasets such as DRIVE [42], SCES [6], and STARE [14] have been widely reused for vessel segmentation and general fundus analysis, while more recent efforts including MURED [39], ROSE [36], and RetinaMix [16] have emphasized diverse imaging modalities and fair demographic representation.

As shown in these resources, current AI research in glaucoma predominantly focuses on computer vision tasks, employing convolutional neural networks (CNNs) [24]–[26] and their derivatives such as U-Net [40] for applications including pathological feature detection [18], [32], [33], optic disc segmentation [43], and three-dimensional structural reconstruction [27]. These approaches have demonstrated strong capability in extracting low-level visual representations such as retinal layer thickness patterns [36], [41], yet they remain limited in associating these imaging findings with clinically interpretable biomarkers, including intraocular pressure dynamics and optic nerve head deformation [23]. Although the combination of multimodal data and clinical reasoning represents a promising frontier, the domain of automated diagnostic report generation, particularly approaches integrating natural language processing corpora with vision-language or large language models [1], [2], [10], [15], [29], [30], remains largely unexplored within glaucoma-oriented artificial intelligence research. This research gap highlights the necessity for next-generation frameworks capable of unifying imaging biomarkers and textual medical reasoning to produce comprehensive, interpretable, and clinically aligned diagnostic outputs.

B. Medical Report Generation for Glaucoma

Automated medical report generation has recently emerged as a transformative paradigm for converting diagnostic imaging data into structured clinical narratives. This approach offers significant potential for reducing physician workload, enhancing reporting consistency, and improving interpretability in clinical decision-making [50]. Substantial progress has been achieved in radiological imaging, particularly in the automatic interpretation of chest X-rays [2]. However, comparable advancements in ophthalmology, and glaucoma in particular, remain considerably limited. Existing studies in other medical domains often employ hybrid architectures that combine computer vision backbones such as convolutional neural networks (CNNs) with sequential language models including recurrent neural networks (RNNs) or long short-term memory (LSTM) models [7], [49]. More recent approaches have extended this framework by introducing vision-language models that align visual representations with textual semantics to generate coherent diagnostic reports.

Applying such methodologies to glaucoma analysis introduces several unique challenges. First, publicly available datasets that pair fundus or OCT images with expert-annotated clinical reports are scarce, restricting the development of data-driven multimodal systems [17], [19]. Second, most existing architectures overlook critical glaucoma-specific biomarkers, such as cup-to-disc ratio, retinal nerve fiber layer (RNFL) thickness, and peripapillary vessel density, which are essential for accurate disease characterization. Third, glaucoma is inherently a progressive disorder that evolves slowly over time, yet current report generation frameworks rarely incorporate temporal modeling or longitudinal progression patterns. These limitations collectively underscore the necessity for specialized multimodal frameworks capable of integrating imaging biomarkers, clinical domain knowledge, and temporal dynamics to generate accurate and clinically meaningful glaucoma diagnostic reports.

III. METHODOLOGICAL FRAMEWORK

As illustrated in Fig. 1, DA-SPL follows an encoder-decoder paradigm with three key components:

- *Dual-Attention Mechanism (DAM)*
- *Parallel-LSTM Network (PLN)*
- *Label Enhancement Module (LEM)*

Each component is described in detail below, followed by the formulation of our optimization objective.

A. Dual-Attention Mechanism

Our encoder begins by partitioning the input fundus image into non-overlapping patches, which are subsequently processed through gated positional self-attention (GPSA) layers [4], [8]. These layers incorporate positional embeddings to maintain spatial relationships within the retinal structure. The feedforward network (FFN) component [47] employs a two-layer architecture with Gaussian Error Linear Unit (GeLU) activation [12] for nonlinear transformation. Following the final GPSA layer, we append a classification token and project

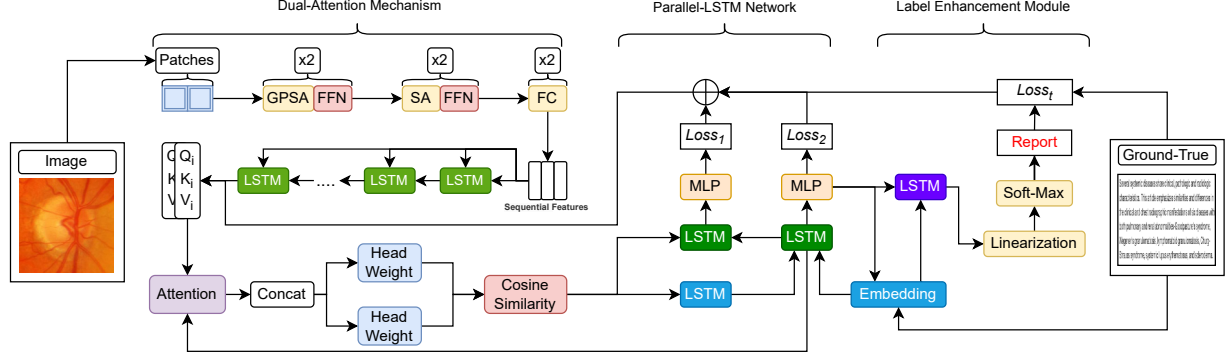


Fig. 1: The Architecture of DA-SPL

the features through a fully connected layer to obtain 1D sequential representations [4], [11], [24]. These features are then processed by the initial LSTM unit [13], [50], coupled with an attention mechanism for iterative refinement before transformer-based processing [47].

The conventional multi-head attention mechanism is enhanced through our dual-weighting approach. The standard multi-head attention computation is given by:

$$att^i_a = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (1)$$

$$att_a = Concat(att_a^1, att_a^2, \dots, att_a^N)W \quad (2)$$

where N is the number of heads, and d_k the scaling factor.

For the first weight: learnable head weights w_a are updated each iteration via

$$w_{a(i)} = softmax(w_{a(i-1)})N \quad (3)$$

and applied to the attention output:

$$att_{wa(i)} = w_{a(i)} * att_{a(i)} \quad (4)$$

Initial weights are uniform; training assigns higher values to heads with greater task relevance.

For the second weight, to enhance salient heads, we compute cosine similarity with the most important head in the batch:

$$cos(i)^j = cos\left(att_{a(i-1)}^j, att_{a(i-1)}^{base}\right) \quad (5)$$

Compared with the single-weight configuration, as demonstrated in TABLE I, the dual-weight implementation exhibits a twofold increase in parameter magnitude. This enhanced weighting scheme within the attention mechanism enables more robust feature emphasis, facilitates multi-scale information aggregation, and yields superior representational capacity while maintaining model generalizability.

Averaging over the batch yields:

$$w_{cos(i)}^j = \frac{\sum_{k \in i} cos(i)_k^j}{N} \quad (6)$$

TABLE I: Comparison of Single Weight & Double Weight

Fold	Weight	Head							
		1	2	3	4	5	6	7	8
1	single	2.737	7.981	2.737	2.737	2.737	2.737	2.737	2.737
	double	5.417	14.964	5.444	5.419	5.508	5.501	5.407	5.425
2	single	7.981	2.737	2.737	2.737	2.737	2.737	2.737	2.737
	double	14.964	5.392	5.476	5.599	5.418	5.490	5.417	5.458
3	single	2.737	2.737	2.737	2.737	2.737	2.737	7.981	2.737
	double	5.511	5.473	5.452	5.425	5.523	5.444	14.964	5.512
4	single	2.737	2.737	2.737	2.737	2.737	2.737	2.737	2.737
	double	5.425	5.376	5.455	5.449	5.392	14.964	5.388	5.338
5	single	7.981	2.737	2.737	2.737	2.737	2.737	2.737	2.737
	double	14.964	5.436	5.573	5.399	5.415	5.466	5.484	5.506
6	single	2.737	2.737	2.737	2.737	2.737	2.737	2.737	2.737
	double	5.526	5.575	5.525	5.506	5.570	5.486	5.558	14.964
7	single	2.737	2.737	2.737	2.737	2.737	2.737	7.981	2.737
	double	5.567	5.519	5.489	5.431	5.531	5.524	14.964	5.486
8	single	2.737	2.737	2.737	2.737	2.737	2.737	2.737	2.737
	double	5.464	5.474	5.495	5.514	14.964	5.542	5.480	5.495
9	single	2.737	2.737	2.737	7.981	2.737	2.737	2.737	2.737
	double	5.422	5.462	5.387	14.964	5.342	5.435	5.445	5.390
10	single	2.737	2.737	7.981	2.737	2.737	2.737	2.737	2.737
	double	5.591	5.502	14.964	5.530	5.523	5.517	5.506	5.592

Blue represents a number without extra units.
Black means there is a unit ($\times 10^{-3}$) after the number.

For the combination of Dual-Weight, We define

$$\beta = \sqrt[n]{\prod_{i=1}^n |w_{cos(i)}|} \quad (7)$$

and adjust the cosine weight by

$$w_{dwa(i)}^j = \beta - w_{cos(i)} \quad (8)$$

This mechanism selectively amplifies attention heads corresponding to pathological samples while attenuating those associated with normal anatomical features. The ReLU-based weight rectification eliminates negative activations, thereby enhancing computational efficiency and sharpening the model's focus on clinically relevant pathological indicators - effectively mimicking the selective attention processes observed in human diagnostic reasoning.

B. Parallel-LSTM Network

The Parallel-LSTM Network consists of three LSTMs but two of them are the same. LSTM (blue, as shown in Fig. 1) is just used to encode features from Dual-Attention Mechanism, as expressed in Eq. 9.

$$h_t^1 = LSTM(T_t^1, w_{dwa(i)}^j * att_{a(i)}^j) \quad (9)$$

where T_t^1 denotes the word embedding of transformer generated. When the first LSTM (dark green) decode the first time, the inputs are T_t^1 , h_t^1 , the word embedding $W_e^2 x_t$ and the hidden state of LSTM, h_{t-1}^2 , as shown in Eq. 10.

$$h_t^2 = LSTM(T_t^1, h_t^1, W_e^2 x_t, h_{t-1}^2) \quad (10)$$

Then the probability of next word can be predicted by a MLP, as appeared in Eq. 11:

$$MLP_{t+1}^1 = softmax(W_{FC1} * h_t^2) \quad (11)$$

Where W_{FC1} is the fully connected layer of MLP. So does the sencond LSTM (dark green), as shown in Eq. 13.

$$h_t^3 = LSTM(T_t^2, h_t^2, W_e^2 x_t) \quad (12)$$

$$MLP_{t+1}^1 = softmax(W_{FC2} * h_t^3) \quad (13)$$

After that, the text features generated by the two MLPs are used to calculate the loss functions and use it in the later stage.

C. Label Enhancement Module

Medical imaging datasets typically comprise disease-specific annotations and clinical descriptors that are critical for multi-label classification tasks. However, limited dataset scales coupled with extensive label spaces often compromise classification precision, subsequently degrading the quality of generated diagnostic reports. To address this, we conceptualize diagnostic labels as report "keywords" and propose a dedicated quality refinement module in DA-SPL that performs post-generation report optimization. Departing from conventional approaches, our novel multi-label classifier operates on the generated report text rather than direct image input. The architecture integrates a category prediction LSTM (illustrated in purple) with a subsequent classification layer, processing the embedded report representation RP_{emb} through the following computational pipeline:

The module first processes RP_{emb} through the category prediction LSTM to generate initial label predictions $Label$, as formalized in Eq.14,

$$Label = softmax(W_t * LSTM(RP_{emb}) + b_t) \quad (14)$$

where W_t and b_t are trainable parameters for flattening the LSTM output. A mapping layer aligns $Label$ with the classification dimensionality, and a softmax yields the probability distribution over all possible tags. This label-focused quality enhancement module refines report accuracy by aligning generated content with diagnostic categories, enabling more precise and clinically relevant medical image report generation.

D. Loss Function

As shown in Eq. 15, $loss$ is the final loss of model, $loss_1$ ($loss_2$) is the cross-entropy loss between reports generated by first (second) LSTM in the Parallel-LSTM Network and true captions. The second LSTM (dark green) generation is confined by an adjustable parameter $\lambda \in (0, 1]$. From the beginning, we set λ to 0.5 at first.

$$loss = loss_1 + \lambda * loss_2 + \alpha * loss_t \quad (15)$$

Since $loss_t$ is merely one-tenth of $loss_1$ and $loss_2$ statistically, $loss_t$ needs to be amplified to balance. Given that $loss_t$ does not directly reflect to reports quality like $loss_1$ and $loss_2$, we set its coefficient $\alpha \in [1, 10)$ at 5. Among them, $loss_t^1$ can be expressed as Eq. 16:

$$loss_t = -\frac{1}{C} * \sum label_i * \log(sig(Label_i)) + (1 - label_i) * \log(1 - sig(Label_i)) \quad (16)$$

Where $label$ represents the true label, $i \in \{0, \dots, n - 1\}$, $label_i \in \{0, 1\}$; n is the number of labels types. $loss_t$ is added in backward propagation of training process for Label Enhancement Module.

IV. DATASET AND IMPLEMENTATION

A. Dataset

As shown in TABLE II, the amount of data has been reduced significantly. For its preprocessing to features like "optic_disc_size" which has specific value (feature, also called factor, structured data), we will refer to the judgment criteria for glaucoma and give "True" or "False". "neuroretinal_rim" is just the description (corpus). "glaucoma_risk_assessment" and "confidence_level" are determined by human subjective judgment, akin to labels, indicating whether the case is classified as glaucoma and providing an individual's evaluation of the judgment. Dataset with its images is available through this link².

TABLE II: One Sample from Public Dataset

Fundus_Features	Ground True
optic_disc_size	large
cup_to_disc_ratio	0.8
isnt_rule_followed	false
rim_pallor	true
rim_color	pale
bayoneting	true
sharp_edge	true
laminar_dot_sign	true
notching	true
rim_thinning	true
additional_observations	null
neuroretinal_rim	Description
glaucoma_risk_assessment	high risk
confidence_level	0.9

B. Implementation

1) *Hyperparameter*: we utilizes a ConViT [4], pretrained on ImageNet [5], with the final classification layer removed to extract 512-dimensional visual feature representations. Both word embeddings and LSTM hidden states are configured with 512-dimensional vectors [13], [50]. Model optimization employs the ADAM optimizer [31] with weight decay. The baseline architecture combines ConViT with standard multi-head attention [47], trained with an initial learning rate of

¹MultiLabel- SoftMarginLoss: <https://pytorch.org/docs>

²https://huggingface.co/datasets/AswanthCManoj/glaucoma_diagnosis_json_analysis

0.0004. Through empirical validation, we set the trade-off parameter $\lambda = 0.5$ for optimal performance. Text generation utilizes beam search with width 5, and automated metric computation is performed using the standard image captioning evaluation toolkit.

2) *Evaluation Metrics*: we assess model performance using three established automatic evaluation metrics for natural language generation: BLEU [38], CIDEr [48], and ROUGE-L [28]. The BLEU metric (B-n, where n denotes n-gram order) calculates modified n-gram precision with a brevity penalty to penalize truncated outputs. CIDEr (CID) employs TF-IDF weighted n-gram matching to emphasize clinically relevant terminology. ROUGE-L (ROU) evaluates the longest common subsequence to measure content coverage. This comprehensive metric suite evaluates report fluency, clinical relevance, and semantic alignment with reference standards. In result tables, the top-performing values are highlighted in bold, while sub-optimal results are underlined.

3) *Computational Environment*: the experimental platform consists of a GPU cluster with the following specifications: $2 \times$ NVIDIA Tesla V100 GPUs (32GB VRAM each), 64GB system RAM, 8-core CPUs per compute node, and a total of 6 interconnected nodes. All experiments were conducted using PyTorch with mixed-precision training enabled.

V. EXPERIMENTAL RESULTS

A. Experimental Configuration

1) *Comparative Model*: we benchmark against five state-of-the-art medical report generation architectures: CNN-RNN [49], CNN-LSTM [7], CaptionNet [50], AoANet [22], and JE-TriNet [51]. All comparative models were hyperparameter-optimized using Optuna³ with Bayesian optimization. Each model underwent rigorous training for 200 epochs, with performance evaluation conducted through ten-fold cross-validation to ensure statistical reliability and generalization capability. The validation protocol maintains identical data splits across all compared methods.

2) *Large Language Model*: to enhance medical report generation, we incorporate both open-source (LlaMA family [2], [45], [46]) and proprietary (GPT series [1], [37]) large language models, fine-tuned for the downstream diagnostic task. The LlaMA models (3.1-70B and 3.1-405B) were deployed via the LlaMA-API platform, while GPT-4O and GPT-O1 were accessed through their official APIs. For visual feature extraction, we employ a CNN backbone (ResNet-152 [11], [24]), whose final fully-connected layers are projected into the LLM's embedding space. The fine-tuning process, detailed in TABLE III, involves: (1) augmenting the LLM with a task-specific vision adapter head, (2) joint optimization of both components using AdamW [31] with cosine learning rate decay (10% warm-up), and (3) cross-entropy loss minimization⁴. This approach enables the LLM to develop task-specific representations while preserving its generative capabilities.

³<https://optuna.org/>

⁴<https://docs.pytorch.org/docs/stable/generated/torch.nn.CrossEntropyLoss>

TABLE III: Fine-tuning with Vision Encoder Head

LLM	Version	DR	LR	BS	MSL	Epoch	GC
GPT	4o	5×10^{-5}	5×10^{-5}	8	512	50	1.0
	o1	5×10^{-5}	5×10^{-5}	8	512	50	1.0
	o1-mini	7×10^{-5}	7×10^{-5}	8	512	50	1.0
	4.1	5×10^{-5}	5×10^{-5}	8	512	50	1.0
LlaMA	3.1-405B	5×10^{-5}	5×10^{-5}	8	512	50	1.0
	3.1-70B	5×10^{-5}	5×10^{-5}	8	512	50	1.0
	3.1-8B	7×10^{-5}	7×10^{-5}	8	512	50	1.0

Note: Dropout Rate = DR; Learning Rate = LR; Batch Size = BS; Max Sequence Length = MSL; Gradient Clipping = GC and Gradient Clipping = GC.

B. Comparative Experiment

Table IV reports the performance of all models on the glaucoma report generation task. While JE-TriNet [51] yields the strongest baseline results and GPT-4o [1] leads among LLMs, DA-SPL achieves the highest overall performance, with notable gains in BLEU-1, BLEU-2, BLEU-4, ROUGE-L, and CIDEr, and a competitive BLEU-3, underscoring its effectiveness in producing accurate and semantically comprehensive reports.

TABLE IV: Comparative Experiment Among Models ($\times 100\%$)

Baseline Model		B-1	B-2	B-3	B-4	ROU	CID
CNN-RNN	Optuna	42.13	26.34	17.25	11.28	36.27	32.41
CNN-LSTM	Optuna	46.52	29.18	19.46	13.47	38.49	37.95
CaptionNet	Optuna	49.27	31.56	21.38	14.52	40.23	45.28
AoANet	Optuna	52.79	33.82	23.74	16.25	43.56	56.31
JE-TriNet	Optuna	<u>54.38</u>	<u>36.17</u>	25.49	<u>17.93</u>	<u>44.87</u>	<u>61.84</u>
LLMs		Encoder					
LlaMa3.1-8B	CNN	50.28	33.69	22.95	16.24	42.85	57.86
LlaMa3.1-70B	CNN	51.72	34.98	24.07	17.15	43.79	59.64
LlaMa3.1-405B	CNN	52.36	35.51	24.59	17.53	44.12	60.48
GPT-4o	CNN	52.94	35.98	24.97	17.86	44.45	61.23
GPT-o1	CNN	52.05	35.27	24.31	17.39	43.92	60.11
GPT-o1-mini	CNN	50.63	33.98	23.14	16.42	43.06	58.37
GPT-4.1	CNN	52.68	35.79	24.78	17.72	44.31	60.94
DA-SPL		56.97	36.71	25.14	18.74	45.08	62.44

C. Ablation Experiment

1) *Proposed Component*: ablation experiments were conducted to evaluate the contribution of each proposed component. As shown in TABLE V, performance improves steadily as modules are added, with LEM providing the highest single-module gain, followed closely by PLN and DAM. Applying hyperparameter tuning via Optuna further enhances results across all configurations. DA-SPL with all modules and tuning achieves the best overall performance, exceeding the untuned configuration by approximately two percentage points across the evaluation metrics.

2) *Backbone & Weight*: TABLE VI presents the ablation results on backbone selection and attention weight strategy. Within the DAM setting, replacing the standard single-weight attention with the dual-weight variant yields slight but consistent improvements across most metrics for the ViT backbone. When comparing backbones, ConViT outperforms ViT under the same attention configuration. Furthermore, for the ConViT backbone, adopting the dual-weight design in the proposed

TABLE V: Ablation Study on Proposed Module ($\times 100\%$)

Module	B-1	B-2	B-3	B4	ROU	CID
+ DAM	10.42	6.85	4.02	2.91	8.64	14.73
+ DAM (Optuna)	24.36	15.28	9.83	7.12	20.17	28.45
+ PLN	31.59	20.47	13.26	9.54	26.83	37.68
+ PLN (Optuna)	39.74	25.83	16.87	12.16	32.74	45.92
+ LEM	46.12	30.21	19.83	14.23	37.89	53.15
+ LEM (Optuna)	52.35	34.68	23.15	16.72	42.36	58.79
+ all	<u>55.12</u>	<u>35.89</u>	<u>24.52</u>	<u>18.12</u>	<u>44.63</u>	<u>61.02</u>
+ all (Optuna)	56.97	36.71	25.14	18.74	45.08	62.44

DA-SPL leads to the highest performance across all metrics, confirming the effectiveness of both the backbone choice and the dual-weight attention mechanism.

TABLE VI: Ablation Study on Backbone & Weight ($\times 100\%$)

DAM Backbone	B-1	B-2	B-3	B4	ROU	CID
ViT (DAM)	54.82	35.12	23.64	17.25	43.72	60.18
ViT (Dual-weight)	54.95	35.27	23.79	17.39	<u>44.25</u>	<u>60.36</u>
Weight (ConViT)	B-1	B-2	B-3	B4	ROU	CID
Single-weight	<u>55.96</u>	<u>35.84</u>	<u>24.21</u>	<u>17.88</u>	43.85	<u>61.07</u>
DA-SPL	56.97	36.71	25.14	18.74	45.08	62.44

3) *Loss Function*: we set $\alpha = 5$ for optimal performance, as determined by an ablation in which we varied $\alpha \in [1, 10]$. As shown in TABLE VII, ablation experiments varying α from 1 to 10 reveal a clear performance peak at $\alpha = 5$, where most key metrics reach their highest or near-highest values. This suggests that $\alpha = 5$ provides an optimal balance between the relative contributions in the loss function, enabling the model to effectively learn salient features without overemphasizing or underweighting specific components.

TABLE VII: Impact of α on Model Performance ($\times 100\%$)

α	B-1	B-2	B-3	B4	ROU	CID
1	38.42	25.31	16.85	11.94	29.78	28.63
2	42.57	28.46	18.92	13.02	33.14	31.25
3	51.86	33.47	22.75	16.59	41.03	55.26
4	55.62	35.94	24.89	18.33	45.08	61.08
5	56.97	36.71	25.14	18.74	45.08	<u>62.44</u>
6	56.21	36.98	24.96	18.59	<u>44.95</u>	61.92
7	54.18	35.26	24.37	18.02	44.01	60.79
8	40.25	27.14	18.05	12.41	31.92	29.87
9	36.83	24.95	16.47	11.08	28.65	27.53
10	33.94	22.83	15.26	10.27	26.48	25.39

4) *Input Modality*: Table VIII reports the impact of different input modality combinations on DA-SPL performance. Here, Factor denotes the inclusion of structured data, such as key clinical indicators extracted from electronic health records. Results show that introducing structured data alongside image and corpus inputs consistently improves performance across all metrics. Notably, the combination of all three modalities, image, corpus, and structured factors, achieves the highest scores, highlighting the complementary role of structured data in enhancing both accuracy and completeness of generated glaucoma diagnostic reports.

D. Supplementary Experiment

Table IX compares the original models with their improved counterparts after incorporating the proposed weight balance

TABLE VIII: Impact of Input Modality ($\times 100\%$)

Data Category			DA-SPL					
Image	Corpus	Factor	B-1	B-2	B-3	B4	ROU	CID
$\times$	$\times$	$\times$	-	-	-	-	-	-
$\times$	$\times$	$\checkmark$	5.12	4.96	5.08	5.21	4.87	5.03
$\times$	$\checkmark$	$\times$	15.20	10.35	7.14	6.23	14.26	20.35
$\times$	$\checkmark$	$\checkmark$	25.45	16.82	11.97	8.92	22.73	31.84
$\checkmark$	$\times$	$\times$	35.68	23.21	16.43	12.25	31.42	43.72
$\checkmark$	$\times$	$\checkmark$	<u>51.34</u>	<u>33.42</u>	<u>23.56</u>	<u>17.42</u>	<u>42.86</u>	<u>58.03</u>
$\checkmark$	$\checkmark$	$\times$	45.12	29.54	20.89	15.63	38.92	52.18
$\checkmark$	$\checkmark$	$\checkmark$	56.97	36.71	25.14	18.74	45.08	62.44

function (Eq. 7). Across all architectures, the enhanced versions ($\checkmark$) achieve consistent improvements over their base-lines ($\times$) in BLEU, ROUGE-L, and CIDEr scores. The gains are most pronounced for transformer-based and attention-augmented models, such as AoANet and JE-TriNet, indicating that the weight balance function effectively optimizes feature contributions within these architectures. Although the performance of JE-TriNet (plus) remains slightly below that of DA-SPL, the observed improvements confirm the broad applicability and effectiveness of the proposed method across different model backbones.

TABLE IX: Comparison of Models & Improved Ones ($\times 100\%$)

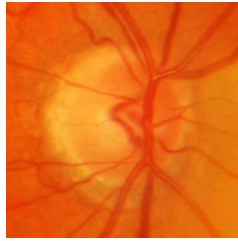
Model	Ver.	B-1	B-2	B-3	B4	ROU	CID
CNN-RNN	$\times$	42.13	26.34	17.25	11.28	36.27	32.41
CNN-RNN	$\checkmark$	43.58	27.46	17.92	11.85	37.42	33.76
CNN-LSTM	$\times$	46.52	29.18	19.46	13.47	38.49	37.95
CNN-LSTM	$\checkmark$	47.96	30.27	20.12	14.03	39.64	39.24
CaptionNet	$\times$	49.27	31.56	21.38	14.52	40.23	45.28
CaptionNet	$\checkmark$	50.63	32.64	22.04	15.08	41.38	46.73
AoANet	$\times$	52.79	33.82	23.74	16.25	43.56	56.31
AoANet	$\checkmark$	54.21	34.91	24.38	16.81	44.71	57.82
JE-TriNet	$\times$	54.38	36.17	25.49	17.93	44.87	61.84
JE-TriNet	$\checkmark$	55.46	36.54	25.88	18.32	45.03	62.10

E. Visualization

Fig. 2 presents two representative case studies from our report generation system. The first case (Panel a) demonstrates a high-risk glaucoma presentation, exhibiting characteristic findings including an enlarged optic disc (cup-to-disc ratio 0.8), diffuse neuroretinal rim thinning (most prominent in inferior and superior quadrants), ISNT rule violation, rim pallor, and associated structural changes. The second case (Panel b) illustrates a healthy control, displaying preserved neuroretinal rim architecture, absence of pathological disc features, and high confidence (95%) in low-risk classification. Comparative analysis reveals that DA-SPL generates reports with improved conciseness and simplified sentence structures compared to ground truth annotations, while maintaining diagnostic accuracy. This optimization enhances clinical utility by improving readability and facilitating rapid interpretation.

VI. CONCLUSION

We present DA-SPL, an innovative dual-attention semantic parallel-LSTM architecture for automated glaucoma diagnostic reporting from fundus imaging. DA-SPL's three key innovations: joint encoder attention mechanisms, parallelized LSTM



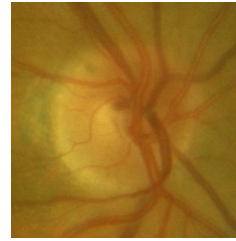
Optic Disc: The optic disc is large, with a cup-to-disc ratio of 0.8. The neuroretinal rim is diffusely thinned, most severely in the inferior and superior quadrants, violating the ISNT rule.

Rim Appearance: The rim is pale with marked pallor, accompanied by bayoneting, sharp disc margins, laminar dot sign, notching, and generalized rim thinning.

Additional Findings: No other significant abnormalities detected.

Glaucoma Risk Assessment: High risk for glaucoma.
Confidence Level: 90%.

(a) Glaucoma Report



Optic Disc: The optic disc is large, with a cup-to-disc ratio of 0.8. The neuroretinal rim is diffusely thinned, most severely in the inferior and superior quadrants, violating the ISNT rule.

Rim Appearance: The rim is pale with marked pallor, accompanied by bayoneting, sharp disc margins, laminar dot sign, notching, and generalized rim thinning.

Additional Findings: No other significant abnormalities detected.

Glaucoma Risk Assessment: Very healthy.
Confidence Level: 95%.

(b) Nomal Eye Report

Fig. 2: Medical Report Generation of DA-SPL on Glaucoma & Nomal Eye

decoder structure, and label enhancement module, collectively enable precise capture of subtle pathological features while optimizing semantic coherence and clinical relevance. The accompanying multimodal glaucoma corpus, comprising fundus images, structured clinical features, and expert-curated reports, represents a significant contribution to medical AI research infrastructure. Comprehensive evaluations across multiple benchmarks confirm DA-SPL's superior performance over existing methods in terms of both quantitative metrics and qualitative clinical utility, generating concise yet diagnostically precise reports that closely mirror expert annotations.

VII. FUTURE WORK

Our ongoing research will focus on expanding and standardizing the proposed glaucoma-specific multimodal corpus, with planned public release to support broader scientific investigation. Future directions include:

- Systematic Integration of LLMs to Enhance Diagnostic Reasoning Capability
- Development of Advanced Multimodal Fusion Technique for Improved Feature Representation
- Clinical Validation Studies to Assess Real-world Applicability

We anticipate that combining comprehensive multimodal data resources with cutting-edge AI methodologies will yield transformative advances in glaucoma diagnosis and management, ultimately bridging the gap between computational research and clinical practice.

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